

Notes on Linear Algebra

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2014

1 Vector spaces

The set of complex numbers is defined as $\mathbb{C} = \{a + bi | a, b \in \mathbb{R}\}$. By definition, $i^2 = -1$.

Addition and multiplication are defined as follows:

$$\begin{aligned}(a + bi) + (c + di) &= (a + c) + (b + d)i, \\ (a + bi)(c + di) &= (ac - bd) + (ad + bc)i.\end{aligned}$$

Addition and multiplication of complex numbers is commutative and associative. Multiplication is distributive over addition. There exists an additive identity (0) and a multiplicative identity (1).

For every $z \in \mathbb{C}$ there is a unique w such that $w + z = 0$, which we denote by $-z$. There exists, for every $z \in \mathbb{C}$, with $z \neq 0$, a w such that $zw = 1$, which we denote by $1/z$. Division is defined as follows: $z/w = z(1/w)$. Powers are defined as usual and have the properties $(z^m)^n = z^{mn}$ and $(zw)^m = z^m w^m$.

By convention, $\mathbb{F}$ will denote either $\mathbb{R}$ or $\mathbb{C}$ depending on context. We call $a \in \mathbb{F}$ a scalar.

Throughout these notes, V and W denote vector spaces over $\mathbb{F}$, and both spaces are over the *same* field $\mathbb{F}$ whenever they appear together.

Let $z = a + bi$ be a complex number. Then $\operatorname{Re} z = a$ and $\operatorname{Im} z = b$. The complex conjugate of z is defined as $\bar{z} = a - bi$. The absolute value is defined as $|z| = \sqrt{a^2 + b^2}$. The following properties hold:

$$\begin{aligned}\operatorname{Re}(w + z) &= \operatorname{Re} w + \operatorname{Re} z, \\ \operatorname{Im}(w + z) &= \operatorname{Im} w + \operatorname{Im} z, \\ z + \bar{z} &= 2 \operatorname{Re} z, \\ z - \bar{z} &= 2(\operatorname{Im} z)i, \\ z\bar{z} &= |z|^2, \\ \overline{w + z} &= \bar{w} + \bar{z}, \\ \overline{\bar{z}} &= z, \\ |wz| &= |w||z|.\end{aligned}$$

A vector space is a set V along with operations of addition and scalar multiplication. Addition is a function that assigns an element $u + v \in V$ to each pair $u, v \in V$. Scalar multiplication is a function that assigns an element $av \in V$ for every $a \in \mathbb{F}$ and $v \in V$. A member of a vector space is called a vector. We leave it to the context to determine whether 0 represents a vector or a scalar. The following properties must hold for V to be considered a vector space. V must be closed under addition and scalar multiplication. Addition must be commutative ($u + v = v + u$) and associative ($(u + v) + w = u + (v + w)$), and scalar multiplication must be associative ($(ab)v = a(bv)$). There must be an additive identity $0 \in V$, so that $v + 0 = v$ for every $v \in V$, and every $v \in V$ must have an additive inverse, that is, some $w \in V$ with $v + w = 0$. The scalar 1 must act as a multiplicative identity ($1v = v$ for every $v \in V$). Finally, both distributive properties must hold: $a(u + v) = au + av$ and $(a + b)v = av + bv$ for all $a, b \in \mathbb{F}$ and $u, v \in V$.

There are several examples of vector spaces: $\mathbb{R}^n$ (the set of real n -tuples), $\mathbb{P}(\mathbb{F})$ (the set of polynomial functions with coefficients in $\mathbb{F}$) etc. Particularly interesting is $\mathbb{R}^2$, where addition and scalar multiplication can be interpreted geometrically (arrow addition and scaling).

A subspace U of V is a subset of V that is itself a vector space under the operations inherited from V . Equivalently, a subset $U \subseteq V$ is a subspace if and only if $0 \in U$ and U is closed under addition and scalar multiplication.

We define the sum of vector spaces as follows: $U = U_1 + U_2 + \cdots + U_n = \{u_1 + u_2 + \cdots + u_n \mid u_j \in U_j\}$. It is interesting to note that U is the smallest subspace of V that contains every U_j . We say V is a direct sum of $U_1, \dots, U_n$ if and only if every element of V can be written uniquely as the sum of $u_1 + u_2 + \cdots + u_n$, $u_j \in U_j$, in

which case we say $V = U_1 \oplus \cdots \oplus U_n$. Suppose $V = U_1 + \cdots + U_n$. Then V is a direct sum of $U_1, \dots, U_n$ if and only if there exists only one way of writing 0 as the sum of $u_1 + \cdots + u_n$, $u_j \in U_j$. Also interesting, if $V = U \oplus W$, then $U \cap W = \{0\}$.

2 Finite-Dimensional Vector Spaces

A linear combination of a list of vectors $(v_1, \dots, v_n)$ is a vector of the form $a_1v_1 + \cdots + a_nv_n$, where $a_j \in \mathbb{F}$.

The span of $(v_1, \dots, v_n)$ is the set of all linear combinations using these vectors. The span of any list of vectors in V is a subspace of V . If a list of vectors (finite by definition) spans V , then V is said to be finite dimensional.

A list of vectors $(v_1, \dots, v_n)$ is called linearly independent if $a_1v_1 + \cdots + a_nv_n = 0$ implies $a_j = 0$ for every j . A list of size two is linearly independent iff neither vector is a scalar multiple of the other. If a vector is removed from a linearly independent list, the resulting list is still linearly independent. The empty list has span $\{0\}$.

In a finite dimensional vector space, the length of every list of linearly independent vectors is less than or equal to the length of every spanning list. If $(v_1, \dots, v_m)$ is linearly dependent and $v_1 \neq 0$, then there exists $j \in \{2, \dots, m\}$ such that $v_j \in \text{span}(v_1, \dots, v_{j-1})$ and removing v_j from $(v_1, \dots, v_m)$ results in a list with the same span. If, for every positive integer n , there exists a linearly independent list of n vectors in V , then V is infinite dimensional. Every subspace of a finite dimensional space is finite dimensional.

A basis for a vector space V is a list of vectors in V that is linearly independent and spans V .

A list $(v_1, \dots, v_n)$ of vectors in V is a basis of V iff every $v \in V$ can be written uniquely in the form

$$v = a_1v_1 + \cdots + a_nv_n.$$

Every spanning list in a vector space can be reduced to a basis of the vector space. Every finite dimensional space has a basis.

If V is finite dimensional, any linearly independent list of vectors in V can be extended to a basis of V .

If V is a finite dimensional vector space and U is a subspace of V , then there is a subspace W of V such that $V = U \oplus W$.

Any two bases of a finite-dimensional vector space have the same length, which is called the dimension $\dim V$ of V .

If V is finite dimensional and U is a subspace of V , then $\dim U \leq \dim V$.

Every spanning list of V with length $\dim V$ is a basis of V . Every linearly independent list with $\dim V$ vectors in V is a basis of V .

If U_1 and U_2 are two subspaces of a finite dimensional space, then

$$\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2).$$

Suppose V is finite dimensional and $U_1, \dots, U_m$ are subspaces of V such that

$$\begin{aligned} V &= U_1 + \cdots + U_m, \\ \dim V &= \dim U_1 + \cdots + \dim U_m. \end{aligned}$$

In that case, $V = U_1 \oplus \cdots \oplus U_m$.

3 Linear Maps

A linear map from V to W is a function $T : V \rightarrow W$ with the following properties:

$$\begin{aligned} T(u + v) &= T(u) + T(v), \text{ for all } u, v \in V, \\ T(av) &= aT(v), \text{ for } a \in \mathbb{F}. \end{aligned}$$

We denote by $\mathbb{L}(V, W)$ the set of linear maps from V to W . Two important linear maps are the map 0, defined as $0(v) = 0$ (on the left side, 0 denotes a linear map, on the right side, 0 denotes a vector) and, when $W = V$, the identity map I , defined as $I(v) = v$.

Suppose that $(v_1, \dots, v_n)$ is a basis of V and $T \in \mathbb{L}(V, W)$. Then, by the linearity of T , if $v = a_1v_1 + \cdots + a_nv_n$ we have $T(v) = a_1T(v_1) + \cdots + a_nT(v_n)$. This implies that the values $T(v_j) \in W$ define a linear transformation completely.

Let $S, T \in \mathbb{L}(V, W)$. We can define addition and scalar multiplication on $\mathbb{L}(V, W)$ as follows:

$$\begin{aligned}(S + T)(v) &= S(v) + T(v), \\ (aT)(v) &= aT(v).\end{aligned}$$

Thus, it can easily be shown that $\mathbb{L}(V, W)$ is a vector space. We can also define the multiplication ST to be the composition of functions: $(ST)(v) = S(T(v))$. It can be shown that this operation is associative, distributive, and has the identity map as a multiplicative identity ($TI = IT = T$).

We say the null space of a linear transformation $T : V \rightarrow W$ is the set $\text{null } T = \{v \in V | T(v) = 0\}$. It can be shown that $\text{null } T$ is a subspace of V . We say T is injective whenever $T(u) = T(v)$ implies $u = v$. T is also injective if and only if $\text{null } T = \{0\}$.

We say the range of a linear transformation $T : V \rightarrow W$ is the set $\text{range } T = \{T(v) | v \in V\}$. It can be shown that $\text{range } T$ is a subspace of W . We say T is surjective whenever $\text{range } T = W$.

If V is a finite dimensional vector space and $T \in \mathbb{L}(V, W)$, then $\text{range } T$ is a finite dimensional subspace of W and $\dim V = \dim \text{null } T + \dim \text{range } T$.

If V and W are finite dimensional vector spaces and $\dim V > \dim W$ then no linear map from V to W is injective. Also, if $\dim V < \dim W$, no linear map from V to W is surjective.

Let $T : \mathbb{F}^n \rightarrow \mathbb{F}^m$ be defined as

$$T(x) = T(x_1, \dots, x_n) = \left(\sum_{k=1}^n a_{1,k}x_k, \dots, \sum_{k=1}^n a_{m,k}x_k \right).$$

Using the statements above, it can be shown that $T(x) = 0$ has more than one solution when $n > m$ (more variables than equations). Also if $n < m$ (more equations than variables), then, for any choice of the coefficients $a_{i,j}$, there is some c for which no x satisfies $T(x) = c$. This relates linear maps to systems of linear equations.

An m -by- n matrix M is a rectangular array with m rows and n columns that can be represented as

$$M = \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{bmatrix}.$$

Suppose $(v_1, \dots, v_n)$ is a basis of V and $(w_1, \dots, w_m)$ is a basis of W . We can write $T(v_k) = a_{1,k}w_1 + \dots + a_{m,k}w_m$ for every $k = 1, \dots, n$ and scalars $a_{j,k}$ for $j = 1, \dots, m$. The m -by- n matrix formed by the a 's is called the matrix of T with respect to the basis $(v_1, \dots, v_n)$ and $(w_1, \dots, w_m)$, which we denote by $M(T, (v_1, \dots, v_n), (w_1, \dots, w_m))$, or $M(T)$ when the context makes the basis clear. The values in column k are the scalars that need to be multiplied by $w_1, \dots, w_m$ to obtain $T(v_k)$.

Unless stated otherwise, if T is a linear map from $\mathbb{F}^n$ to $\mathbb{F}^m$, the basis is the standard one (where the k -th vector in the basis has 1 in the k -th slot and 0 in every other slot).

Define matrix addition and scalar multiplication to be element-wise. The set $\text{Mat}(m, n, \mathbb{F})$ contains all m -by- n matrices over $\mathbb{F}$. It is also a vector space.

If $M(T)$ is a m -by- n matrix that can be indexed by a 's and $M(S)$ is a n -by- p matrix that can be indexed by b 's, then we define the m -by- p matrix $M(T)M(S)$ (indexed by c 's) to have the following elements:

$$c_{j,k} = \sum_{r=1}^n a_{j,r}b_{r,k}.$$

Using this definition of matrix multiplication, $M(T)M(S) = M(TS)$.

Suppose $(v_1, \dots, v_n)$ is a basis of V . If $v \in V$, then there exist unique scalars $b_1, \dots, b_n$ such that $v = b_1v_1 + \dots + b_nv_n$. We denote the matrix of v as

$$M(v) = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}.$$

Using this definition, $M(T(v)) = M(T)M(v)$.

We say a linear map $T \in \mathbb{L}(V, W)$ is invertible if there exists a linear map S such that $ST = I$ (identity map on V) and $TS = I$ (identity map on W). If T is invertible, its inverse is unique and denoted by T^{-1} . A linear

map is invertible if and only if it is injective and surjective. Two vector spaces are called isomorphic if there is an invertible linear map between them.

Two finite-dimensional vector spaces over the same field $\mathbb{F}$ are isomorphic if and only if they have the same dimension. Thus, every finite-dimensional vector space over $\mathbb{F}$ is isomorphic to some $\mathbb{F}^n$.

Suppose $(v_1, \dots, v_n)$ is a basis of V and $(w_1, \dots, w_m)$ is a basis of W . Then M is an invertible map between $\mathbb{L}(V, W)$ and $\text{Mat}(m, n, \mathbb{F})$. In other words, M is a one-to-one correspondence between linear maps in $\mathbb{L}(V, W)$ and m -by- n matrices over $\mathbb{F}$.

If V and W are finite dimensional, then $\mathbb{L}(V, W)$ is finite dimensional and $\dim \mathbb{L}(V, W) = (\dim V)(\dim W)$.

A linear map from a vector space V to itself is called an operator; the set of operators on V is denoted by $\mathbb{L}(V)$. If V is finite dimensional and $T \in \mathbb{L}(V)$, the following are equivalent: T is injective, T is surjective, T is invertible.

4 Polynomials

A function $p : \mathbb{F} \rightarrow \mathbb{F}$ is called a polynomial with coefficients in $\mathbb{F}$ if there exist $a_0, \dots, a_m \in \mathbb{F}$ such that

$$p(z) = a_0 + a_1 z + \dots + a_m z^m,$$

for all $z \in \mathbb{F}$. If p can be written in the form above and $a_m \neq 0$, we say p has degree m . We use $\mathbb{P}(\mathbb{F})$ to denote the vector space of all polynomials and $\mathbb{P}_m(\mathbb{F})$ to denote the vector space of polynomials with degree at most m . A number $\lambda \in \mathbb{F}$ is called a root of p if $p(\lambda) = 0$. The polynomial $p(z) = 0$ has degree $-\infty$. If $p \in \mathbb{P}(\mathbb{F})$ is a polynomial with degree $m \geq 1$, λ is a root of p if and only if there is $q \in \mathbb{P}(\mathbb{F})$ with degree $m - 1$ such that $p(z) = (z - \lambda)q(z)$ for all $z \in \mathbb{F}$.

A polynomial with degree $m \geq 0$ has at most m distinct roots in $\mathbb{F}$. Suppose $p(z) = a_0 + a_1 z + \dots + a_m z^m = 0$ for every $z \in \mathbb{F}$, then $a_0 = \dots = a_m = 0$. This implies that $(1, z, \dots, z^m)$ is a basis for $\mathbb{P}_m(\mathbb{F})$.

If $p, q \in \mathbb{P}(\mathbb{F})$ and $p \neq 0$, then there exist unique $s, r \in \mathbb{P}(\mathbb{F})$ such that $q = sp + r$ and $\deg r < \deg p$.

The fundamental theorem of algebra states that every nonconstant polynomial with complex coefficients has a root.

If $p \in \mathbb{P}(\mathbb{C})$ is a nonconstant polynomial, then p has a factorization, unique except for the order of the factors, of the form

$$p(z) = c(z - \lambda_1) \dots (z - \lambda_m),$$

where $c, \lambda_1, \dots, \lambda_m \in \mathbb{C}$.

If p is a polynomial with real coefficients and $\lambda \in \mathbb{C}$ is a root of p , then so is $\bar{\lambda}$.

Let $\alpha, \beta \in \mathbb{R}$. Then there is a polynomial factorization of the form $x^2 + \alpha x + \beta = (x - \lambda_1)(x - \lambda_2)$, with $\lambda_1, \lambda_2 \in \mathbb{R}$, if and only if $\alpha^2 \geq 4\beta$.

If $p \in \mathbb{P}(\mathbb{R})$ is a nonconstant polynomial, then p has a factorization, unique except for the order of the factors, of the form

$$p(x) = c(x - \lambda_1) \dots (x - \lambda_m)(x^2 + \alpha_1 x + \beta_1) \dots (x^2 + \alpha_M x + \beta_M),$$

where $c, \lambda_1, \dots, \lambda_m \in \mathbb{R}$, $(\alpha_1, \beta_1), \dots, (\alpha_M, \beta_M) \in \mathbb{R}^2$ with $\alpha_j^2 < 4\beta_j$ for each j , and m or M may be equal to 0.

For each m with $1 \leq m \leq n$, there exists a polynomial $p \in \mathbb{P}_n(\mathbb{F})$ with exactly m distinct roots in $\mathbb{F}$.

Suppose that $z_1, \dots, z_{m+1}$ are distinct elements of $\mathbb{F}$ and $w_1, \dots, w_{m+1} \in \mathbb{F}$. Then there exists a unique polynomial $p \in \mathbb{P}_m(\mathbb{F})$ such that $p(z_j) = w_j$ for $j = 1, \dots, m + 1$.

If $p \in \mathbb{P}(\mathbb{C})$ has degree m , then p and its derivative p' have no roots in common if and only if p has m distinct roots. Every polynomial with odd degree and real coefficients has a real root.

5 Eigenvalues and Eigenvectors

It is useful to consider the behavior of an operator $T \in \mathbb{L}(V)$ on a subspace U of V . We denote by $T|_U$ the restriction of T to the domain U . We say that U is invariant under T if $T|_U$ is an operator on U ; in other words, if $T(u) \in U$ for every $u \in U$. It is easy to see that $\text{null } T$ and $\text{range } T$ are invariant under T .

A scalar $\lambda \in \mathbb{F}$ is called an eigenvalue of $T \in \mathbb{L}(V)$ if there exists a nonzero vector $u \in V$ such that $T(u) = \lambda u$. Therefore, T has an one dimensional invariant subspace if and only if T has an eigenvalue. Since the equation $T(u) = \lambda u$ is equivalent to $(T - \lambda I)(u) = 0$, λ is an eigenvalue of T if and only if $T - \lambda I$ is not injective (it maps a

nonzero vector to 0). When V is finite dimensional, this is in turn equivalent to $T - \lambda I$ being not surjective, and to $T - \lambda I$ being not invertible.

Suppose $T \in \mathbb{L}(V)$ and $\lambda \in \mathbb{F}$ is an eigenvalue of T . A nonzero vector $u \in V$ is called an eigenvector of T corresponding to λ if $T(u) = \lambda u$. Thus, the set of eigenvectors corresponding to λ , together with 0, is $\text{null}(T - \lambda I)$, a subspace of V .

As an example, consider $T \in \mathbb{L}(\mathbb{F}^2)$, where $T(w, z) = (-z, w)$. If $\mathbb{F} = \mathbb{R}$, T has no eigenvalues. If $\mathbb{F} = \mathbb{C}$, its eigenvalues are i and $-i$.

Let $T \in \mathbb{L}(V)$. Suppose $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues of T and $v_1, \dots, v_m$ are corresponding eigenvectors. Then $(v_1, \dots, v_m)$ is linearly independent. As a corollary, if V is finite dimensional, each operator on V has at most $\dim V$ distinct eigenvalues.

If $T \in \mathbb{L}(V)$, we can denote TT by T^2 . In general, T^m denotes the application of T m times. We let T^0 denote I . If T is invertible, its inverse is denoted by T^{-1} and $T^{-m} = (T^{-1})^m$. From these definitions, it follows that $T^m T^n = T^{m+n}$ and $(T^m)^n = T^{mn}$ for arbitrary integers n and m as long as T is invertible and non-negative integers if T is not invertible.

If $T \in \mathbb{L}(V)$ and $p \in \mathbb{P}(\mathbb{F})$ is a polynomial given by $p(z) = a_0 + a_1 z + \dots + a_m z^m$ for $z \in \mathbb{F}$, we define $p(T)$ as the operator $p(T) = a_0 I + a_1 T + \dots + a_m T^m$. The function from $\mathbb{P}(\mathbb{F})$ to $\mathbb{L}(V)$ is linear. If p and q are polynomials, $(pq)(z) = p(z)q(z)$. It is also true that $(pq)(T) = p(T)q(T)$ for any $T \in \mathbb{L}(V)$.

Every operator on a finite-dimensional, nonzero, complex vector space has an eigenvalue.

If V is a finite-dimensional nonzero complex vector space and $T \in \mathbb{L}(V)$, there exists a basis of V with respect to which the matrix of T has zeros everywhere in the first column except possibly in the first entry.

The diagonal of a square matrix consists of the entries along the straight line from the upper left corner to the bottom right corner $(a_{1,1}, \dots, a_{n,n})$. A matrix is called upper triangular if all the entries below the diagonal are zero (* denotes irrelevant values, 0 represents all the entries below the diagonal):

$$\begin{bmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}.$$

Suppose $T \in \mathbb{L}(V)$ and $(v_1, \dots, v_n)$ is a basis of V . Then the following are equivalent: the matrix of T wrt $(v_1, \dots, v_n)$ is upper triangular; $T(v_k) \in \text{span}(v_1, \dots, v_k)$ for each $k = 1, \dots, n$; $\text{span}(v_1, \dots, v_k)$ is invariant under T for each $k = 1, \dots, n$.

Suppose V is a finite-dimensional nonzero complex vector space and $T \in \mathbb{L}(V)$. Then T has an upper-triangular matrix with respect to some basis of V .

Suppose $T \in \mathbb{L}(V)$ has an upper-triangular matrix wrt some basis of V . Then T is invertible if and only if all the entries on the diagonal of that upper-triangular matrix are nonzero.

Suppose $T \in \mathbb{L}(V)$ has an upper-triangular matrix wrt some basis of V . Then the eigenvalues of T consist precisely of the entries in the diagonal of that upper-triangular matrix.

A diagonal matrix is a square matrix that is 0 everywhere except possibly the diagonal. An operator $T \in \mathbb{L}(V)$ has a diagonal matrix wrt some basis of V if and only if V has a basis consisting of eigenvectors of T . If $T \in \mathbb{L}(V)$ has $\dim V$ distinct eigenvalues, then T has a diagonal matrix wrt some basis of V .

Suppose $T \in \mathbb{L}(V)$. Let $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T . Then the following are equivalent:

- T has a diagonal matrix wrt some basis of V ;
- V has a basis consisting of eigenvectors of T ;
- there exist one-dimensional subspaces $U_1, \dots, U_n$ of V , with $n = \dim V$, each invariant under T , such that $V = U_1 \oplus \dots \oplus U_n$;
- $V = \text{null}(T - \lambda_1 I) \oplus \dots \oplus \text{null}(T - \lambda_m I)$;
- $\dim V = \dim \text{null}(T - \lambda_1 I) + \dots + \dim \text{null}(T - \lambda_m I)$.

Every operator on a finite-dimensional, nonzero, real vector space has an invariant subspace of dimension 1 or 2.

Suppose U and W are subspaces of V with $V = U \oplus W$. Each vector $v \in V$ can be written uniquely in the form $v = u + w$. With this representation, define $P_{U,W} \in \mathbb{L}(V)$ by $P_{U,W}(v) = u$. This is called the projection onto U with null space W .

Every operator on an odd-dimensional real vector space has an eigenvalue.

If $T \in \mathbb{L}(V)$ is invertible and $\lambda \in \mathbb{F} \setminus \{0\}$, then λ is an eigenvalue of T if and only if $\frac{1}{\lambda}$ is an eigenvalue of T^{-1} .
 If V is finite dimensional, then for any $S, T \in \mathbb{L}(V)$ the operators ST and TS have the same eigenvalues.

6 Inner-Product Spaces

The length of a vector x in $\mathbb{R}^2$ or $\mathbb{R}^3$ is called the norm of x and denoted by $\|x\|$. For $x = (x_1, x_2) \in \mathbb{R}^2$, $\|x\| = \sqrt{x_1^2 + x_2^2}$. In general, the norm $\|x\|$ of $x \in \mathbb{R}^n$ equals $\sqrt{x_1^2 + \dots + x_n^2}$.

The dot product of $x, y \in \mathbb{R}^n$, denoted $x \cdot y$ is defined by $x \cdot y = x_1 y_1 + \dots + x_n y_n$. Therefore, $\|x\|^2 = x \cdot x$. For $x, y \in \mathbb{C}^n$ the dot product is defined as $x \cdot y = x_1 \overline{y_1} + \dots + x_n \overline{y_n}$, and the norm is defined as $\|x\| = \sqrt{|x_1|^2 + \dots + |x_n|^2}$. The dot product is a special case of a function called an inner product.

An inner-product on V is a function that takes each pair (u, v) of elements of V to a number $\langle u, v \rangle \in \mathbb{F}$ such that $\langle v, v \rangle \geq 0$ for every v ; $\langle v, v \rangle = 0$ if and only if $v = 0$; $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ for all $u, v, w \in V$; $\langle av, w \rangle = a \langle v, w \rangle$ for all $a \in \mathbb{F}$ and $v, w \in V$; and $\langle v, w \rangle = \overline{\langle w, v \rangle}$ for all $v, w \in V$. Clearly, when $\mathbb{F} = \mathbb{R}$, $\langle v, w \rangle = \langle w, v \rangle$. An inner-product space is a vector space V along with an inner-product on V . The Euclidean inner-product on $\mathbb{F}^n$ is the standard inner-product on $\mathbb{F}^n$ and is defined in the same way as the dot product.

We can define the inner product between polynomials in $\mathbb{P}_m(\mathbb{F})$ in the following way:

$$\langle p, q \rangle = \int_0^1 p(x) \overline{q(x)} dx. \quad (1)$$

From the basic properties of an inner product space V , the following can also be derived for any $a \in \mathbb{F}$ and $v, u, w \in V$: $\langle w, 0 \rangle = \langle 0, w \rangle = 0$; $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$; $\langle u, av \rangle = \overline{a} \langle u, v \rangle$.

The norm of $v \in V$ is given by $\|v\| = \sqrt{\langle v, v \rangle}$. We have that $\|v\| = 0$ iff $v = 0$. Also, $\|av\| = |a| \|v\|$ for any $a \in \mathbb{F}$.

Two vectors u and v are said to be orthogonal if $\langle u, v \rangle = 0$. This also implies that $\langle v, u \rangle = 0$. The vector 0 is orthogonal to every vector, and is the only vector orthogonal to itself.

If u and v are nonzero vectors in $\mathbb{R}^2$, $\langle u, v \rangle = \|u\| \|v\| \cos \theta$, where θ is the angle between the two vectors.

The following properties have important geometrical interpretations in $\mathbb{R}^2$ and $\mathbb{R}^3$.

If $u, v \in V$ are orthogonal, then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$. This is known as the Pythagorean theorem.

For any vector $u \in V$ and a given nonzero vector $v \in V$, we can write u as a scalar multiple of v plus a vector orthogonal to v in the following way:

$$u = \frac{\langle u, v \rangle}{\|v\|^2} v + (u - \frac{\langle u, v \rangle}{\|v\|^2} v).$$

If $u, v \in V$, then $|\langle u, v \rangle| \leq \|u\| \|v\|$, and this is an equality if and only if one of u, v is a scalar multiple of the other. This is known as the Cauchy-Schwarz inequality.

If $u, v \in V$, then $\|u + v\| \leq \|u\| + \|v\|$, and this is an equality if and only if one of u, v is a non-negative multiple of the other. This is known as the Triangle Inequality.

If $u, v \in V$, then $\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2)$. This is known as the parallelogram equality.

A list $(e_1, \dots, e_m)$ of vectors in V is orthonormal if $\langle e_j, e_k \rangle = 0$ when $j \neq k$ and $\langle e_j, e_k \rangle = 1$ when $j = k$.

If $(e_1, \dots, e_m)$ is an orthonormal list of vectors, then $\|a_1 e_1 + \dots + a_m e_m\|^2 = |a_1|^2 + \dots + |a_m|^2$. Thus, every orthonormal list of vectors is linearly independent. An orthonormal basis of V is an orthonormal list of vectors in V that is also a basis of V .

The following is very useful to find the scalars to write any v as a linear combination of an orthonormal basis of V . Suppose $(e_1, \dots, e_n)$ is an orthonormal basis of V . Then $v = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_n \rangle e_n$. Also, $\|v\|^2 = |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_n \rangle|^2$.

The Gram-Schmidt process can be used to find an orthonormal list of vectors given a linearly independent list. If $(v_1, \dots, v_m)$ is a linearly independent list of vectors in V , then there exists an orthonormal list $(e_1, \dots, e_m)$ of vectors in V such that $\text{span}(v_1, \dots, v_j) = \text{span}(e_1, \dots, e_j)$ for $j = 1, \dots, m$. We let $e_1 = v_1 / \|v_1\|$, and e_j for $j > 1$ is given by the following:

$$e_j = \frac{v_j - \langle v_j, e_1 \rangle e_1 - \dots - \langle v_j, e_{j-1} \rangle e_{j-1}}{\|v_j - \langle v_j, e_1 \rangle e_1 - \dots - \langle v_j, e_{j-1} \rangle e_{j-1}\|}.$$

Every finite-dimensional inner-product space has an orthonormal basis. Also, if V is finite dimensional, every orthonormal list of vectors in V can be extended to an orthonormal basis of V .

Suppose $T \in \mathbb{L}(V)$. If T has an upper-triangular matrix with respect to some basis of V , then T has an upper-triangular matrix with respect to some orthonormal basis of V .

If V is a finite-dimensional nonzero complex inner-product space and $T \in \mathbb{L}(V)$, then T has an upper-triangular matrix with respect to some orthonormal basis of V .

If U is a subset of V , the orthogonal complement of U , denoted by $U^\perp$, is the set of vectors that are orthogonal to every vector in U . Thus, $U^\perp = \{v \in V \mid \langle v, u \rangle = 0 \text{ for all } u \in U\}$. It can be shown that $U^\perp$ is always a subspace of V , $V^\perp = \{0\}$, and that $\{0\}^\perp = V$. Also, if $U_1 \subset U_2$, $U_1^\perp \supset U_2^\perp$.

If U is a finite-dimensional subspace of V , then $V = U \oplus U^\perp$, and also $(U^\perp)^\perp = U$.

Let U be a finite-dimensional subspace of V . Then every $v \in V$ can be written uniquely in the form $v = u + w$ for $u \in U$ and $w \in U^\perp$, in which case we say $P_U(v) = u$. The function P_U is an operator on V and also has the following properties: $\text{range } P_U = U$; $\text{null } P_U = U^\perp$, $v - P_U(v) \in U^\perp$ for every $v \in V$; $P_U^2 = P_U$; $\|P_U(v)\| \leq \|v\|$ for every $v \in V$.

It is also important to note that if $(e_1, \dots, e_m)$ is an orthonormal basis of U , $P_U(v) = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m$.

The following proposition has led to many applications of inner-product spaces outside of pure mathematics.

Suppose U is a finite-dimensional subspace of V and $v \in V$. Then $\|v - P_U(v)\| \leq \|v - u\|$ for every $u \in U$. Furthermore, if $u \in U$ and the inequality is an equality, then $u = P_U(v)$. Intuitively, this proposition means that for a given v , the nearest point in U to v is the projection of v onto U .

The general form of this minimization problem is as follows. Consider an inner-product space V and a finite-dimensional subspace U . Given a vector $v \in V$, the objective is to find the nearest vector $u \in U$. The solution consists simply of finding an orthonormal basis for U and projecting v onto U .

In this minimization scenario, it is particularly interesting to consider $C[a, b]$, the (infinite dimensional) vector space composed of continuous functions in an interval $[a, b]$. The subspace $\mathbb{P}_m(\mathbb{F})$ of polynomials of degree at most m is finite dimensional, so the results above apply even though $C[a, b]$ is not. Given the inner product $\langle f, g \rangle = \int_a^b f(x) \overline{g(x)} dx$ (the analogue on $[a, b]$ of Eq. 1, pg. 6), it is easy to find the polynomial in $\mathbb{P}_m(\mathbb{F})$ that best approximates a given continuous function by computing some definite integrals.

A linear functional on V is a linear map from V to the scalars $\mathbb{F}$. For instance, let $\varphi : \mathbb{F}^3 \rightarrow \mathbb{F}$ be defined as $\varphi(z_1, z_2, z_3) = 2z_1 + 3z_2 + 1z_3$, then φ is a linear functional on $\mathbb{F}^3$.

Suppose V is finite dimensional and φ is a linear functional on V . Then there is a unique vector $v \in V$ such that $\varphi(u) = \langle u, v \rangle$ for every $u \in V$.

Let $T \in \mathbb{L}(V, W)$. The adjoint of T , denoted T^* , is defined as follows. Fix $w \in W$. Consider the linear functional that maps $v \in V$ to $\langle Tv, w \rangle$. Let T^*w be the unique vector in V such that this linear functional is given by taking inner products with T^*w . In other words, T^*w is the unique vector in V such that $\langle Tv, w \rangle = \langle v, T^*w \rangle$ for all $v \in V$.

It is possible to show that $T^* \in \mathbb{L}(W, V)$. Adjoints also have the following properties: $(S + T)^* = S^* + T^*$ for all $S, T \in \mathbb{L}(V, W)$; $(aT)^* = \overline{a}T^*$ for all $a \in \mathbb{F}$, $T \in \mathbb{L}(V, W)$; $(T^*)^* = T$ for all $T \in \mathbb{L}(V, W)$; $I^* = I$; $(ST)^* = T^*S^*$. Also, $\text{null } T^* = (\text{range } T)^\perp$ and $\text{range } T^* = (\text{null } T)^\perp$.

The conjugate transpose of a matrix is obtained by exchanging rows and columns and then taking the complex conjugate of each entry. Suppose $T \in \mathbb{L}(V, W)$. If $(e_1, \dots, e_n)$ is an orthonormal basis of V and $(f_1, \dots, f_m)$ is an orthonormal basis of W , then $M(T^*, (f_1, \dots, f_m), (e_1, \dots, e_n))$ is the conjugate transpose of $M(T, (e_1, \dots, e_n), (f_1, \dots, f_m))$.

Let V be finite dimensional and $T \in \mathbb{L}(V)$. A scalar $\lambda \in \mathbb{F}$ is an eigenvalue of T if and only if $\overline{\lambda}$ is an eigenvalue of T^* .

Let $T \in \mathbb{L}(V)$ and U be a subspace of V . Then U is invariant under T if and only if $U^\perp$ is invariant under T^* .

Let $T \in \mathbb{L}(V, W)$. Then T is injective if and only if T^* is surjective, and T is surjective if and only if T^* is injective. Also, $\dim \text{range } T^* = \dim \text{range } T$.

Let A be an m -by- n matrix of real numbers. The dimension of the span of the columns of A (in $\mathbb{R}^m$) equals the dimension of the span of the rows of A (in $\mathbb{R}^n$).

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References

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